

Research with References

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1 Introduction

Machine learning has transformed many fields. Convolutional neural networks achieved breakthrough results in image recognition [1]. Transformer architectures revolutionized natural language processing [2], leading to large language models that can perform complex reasoning tasks [3].

2 Related Work

The challenge of PDF parsing has been studied extensively. Early work focused on rule-based approaches [4], while more recent methods employ deep learning for document layout analysis [5]. Our approach combines traditional heuristics with adaptive thresholds, avoiding the need for training data.

3 Discussion

Despite significant progress, PDF-to-text conversion remains challenging due to the format's low-level representation. As noted by Smith et al. [6], the gap between visual presentation and logical structure is the fundamental obstacle.

References

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